

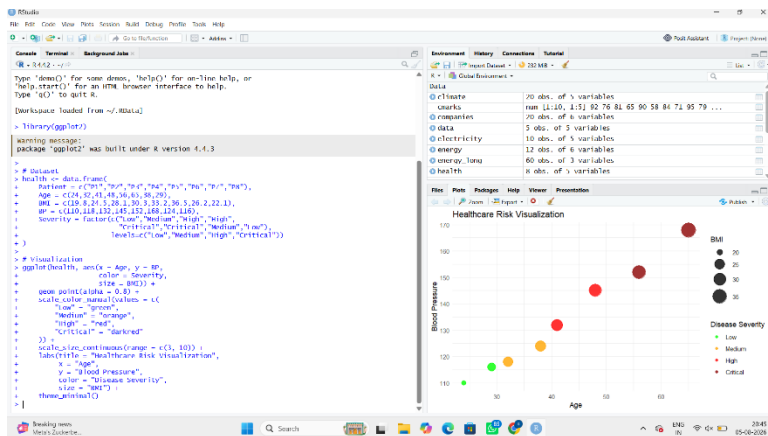
DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 4 (AT1)

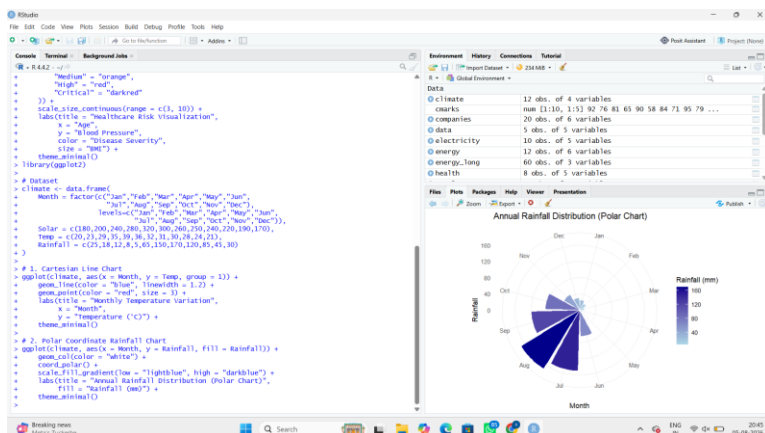
Chart Construction Test

Question 1 (Healthcare Analytics)



- Age → x-axis : identifies risk progression with age.
- Blood Pressure → y-axis : key medical risk indicator.
- Severity → color : traffic-light colors quickly communicate danger.
- BMI → size : larger points emphasize obesity-related risk.

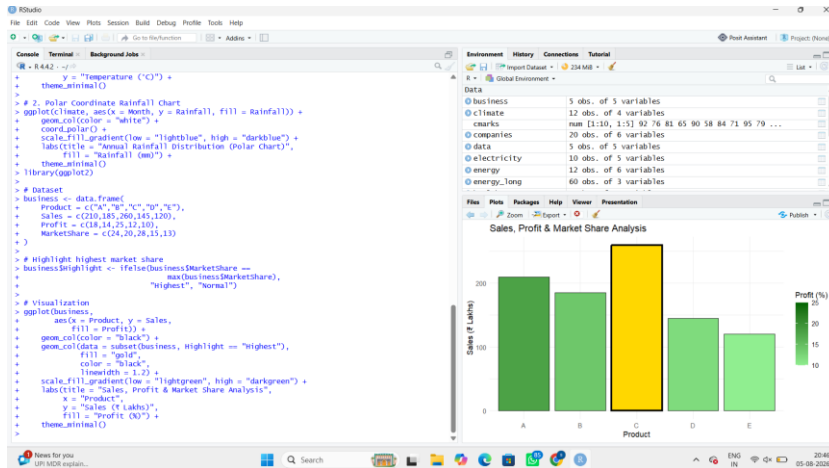
Question 2 (Climate Science)



- Cartesian chart : best for observing temperature trends and exact comparisons.

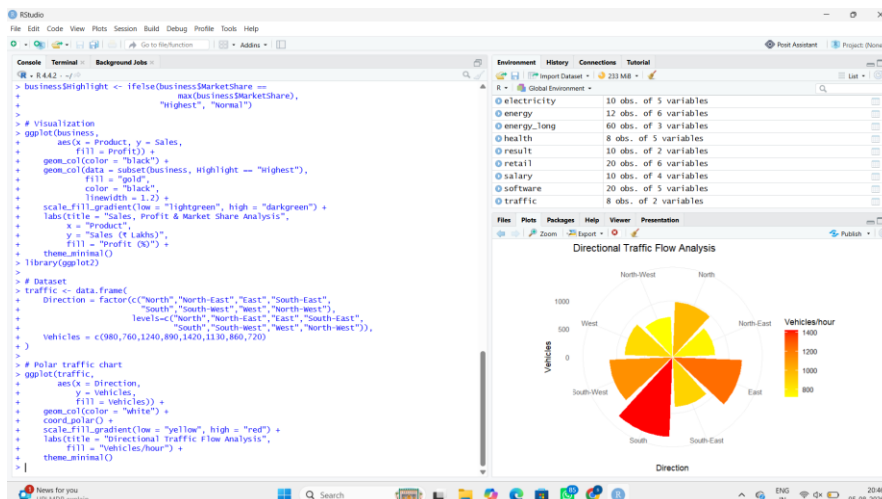
- Polar chart : emphasizes cyclical annual rainfall patterns.
- Limitation of Polar : difficult to compare exact rainfall magnitudes.

Question 3 (Business Analytics & Marketing)



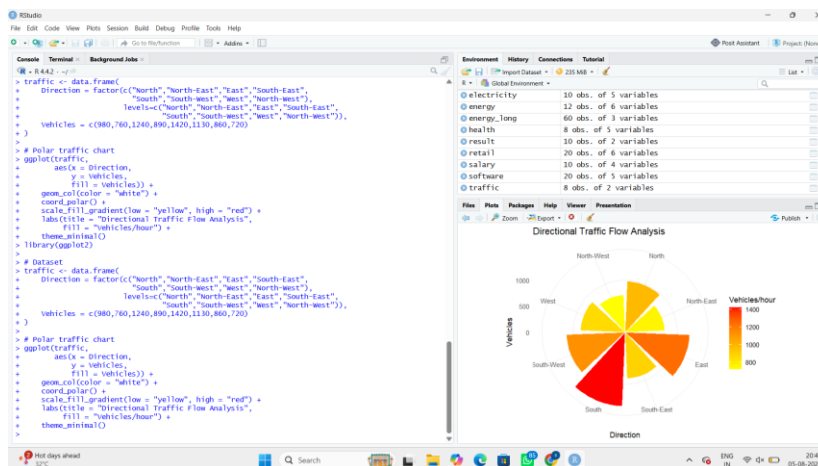
- Light green → dark green naturally represents increasing profitability.
- Gold highlight immediately draws attention to the market leader (Product C).
- Helps managers quickly identify high-sales, high-profit products.

Question 4 (Transportation Engineering)



- Traffic data is directional (360°).
- Polar charts preserve spatial orientation (North, East, South, West).
- Easier to identify traffic-heavy directions, especially South and East.

Question 5 (Environmental Sustainability Dashboard)



- Bar chart : compares AQI values across cities.
- Scatter plot : reveals relationship between green cover and CO₂ emissions.
- Point size : indicates population impact.
- AQI color palette :
 - Green → Good
 - Yellow → Moderate
 - Orange → Poor
 - Red → Very Poor

This dashboard supports environmental monitoring, urban planning, and sustainability decision-making by combining pollution, vegetation, emissions, and population into a single analytical view.

